

Danny Halovanic | Electronic Portfolio

May 07, 2025

Contents

1	Home	5
2	About Me	7
2.1	My Skills	7
3	Projects	9
3.1	Project 1: Comparing Methods for Finding Roots of Polynomials	9
3.2	Project 2: Modeling Prediction of Type II Diabetes in a Population	9
4	Skills	11
4.1	Technical Skills	11
4.2	Other Skills	11
5	Contact	13

Chapter 1

Home

Welcome to my electronic portfolio!

I'm **Danny Halovanic**, an up-and-coming undergraduate with expertise in a wide variety of fields in math and computer science. This portfolio showcases my projects, skills, and experience.

Navigate through the sections to explore:

- **Projects:** Analyses, and reports.
- **Skills:** Programming, statistical methods, and tools.
- **Contact:** How to reach me.

Feel free to explore, and thank you for visiting!

Chapter 2

About Me

Hi, I'm Danny Halovanic, a graduating senior at West Chester University pursuing a double major in Computer Science and Applied and Computational Mathematics.

2.1 My Skills

- **Data Analysis:** Cleaning, transformation, and visualization (R, Python, Tableau)
- **Statistical Modeling:** Regression analysis, hypothesis testing
- **Programming:** R, Python, SQL, Java



Chapter 3

Projects

3.1 Project 1: Comparing Methods for Finding Roots of Polynomials

This report explored and compared the performance of various root-finding techniques—both algebraic and numerical—for polynomials with extreme coefficient magnitudes. The focus was on assessing numerical stability, accuracy, and computational efficiency across methods such as the traditional algebraic formula, low-entropy expressions, as well as numerical methods such as Newton’s and Bisection methods.

[View the report here \(PDF\)](#)

3.2 Project 2: Modeling Prediction of Type II Diabetes in a Population

Using R Markdown and techniques from MAT 325 (Numerical Analysis I), this project modeled the prediction rate of Type II diabetes in the Pima Indian population. Statistical and numerical methods were employed to analyze trends, identify key predictors, and estimate the likelihood of diabetes occurrence based on various health indicators.

[View the project here \(PDF\)](#)

Chapter 4

Skills

4.1 Technical Skills

- R (ggplot2, dplyr, shiny)
- Python (numpy)
- SQL
- Java
- HTML/CSS
- LaTeX

4.2 Other Skills

- Proficiency in Spanish
- Time Management
- Problem Solving
- Communication
- Attention to Detail

Chapter 5

Contact

Email: dan.halovanic01@gmail.com

LinkedIn: [Joseph Halovanic](#)

GitHub: [DannyHal01](#)